

Objective

Write a GNU Radio flowgraph (or pure Python with GNU Radio bindings) that listens to a simulated signal source, measures received signal power, and classifies the signal as analog or digital based on observable properties.

Background

GNU Radio is an open-source software toolkit for building software-defined radio (SDR) systems. In real deployments, SDR receivers capture raw IQ samples from an antenna; signal power and modulation type are key indicators for threat detection and spectrum monitoring. This task uses simulated sources so no hardware is required.

Requirements

1. Create a GNU Radio flowgraph (.grc file or equivalent Python script) that:
 - a. Generates at least two different simulated signals:
 - One analog signal (e.g., FM-modulated audio tone)
 - One digital signal (e.g., BPSK or QPSK with a pseudo-random bit stream)
 - b. Measures the received power of each signal (in dBm or relative units).
 - c. Applies a power threshold: if power exceeds the threshold, flag the signal as "detected".
 - d. Classifies the detected signal as analog or digital using at least one observable property (e.g., bandwidth, spectral shape, amplitude variance).
2. Print or log the results in a readable format:
 - Signal type (analog / digital)
 - Measured power
 - Detection status (detected / not detected)
3. The solution must run end-to-end without any physical SDR hardware.

Deliverables

- A .grc flowgraph file OR a self-contained Python script using GNU Radio
- A short README.md (max 1 page) explaining:
 - How to install dependencies
 - How to run the script
 - What each part of the flowgraph / code does
 - How you chose the classification method

How to Test (without hardware)

1. Install GNU Radio 3.10+ on Linux/macOS, or use the official Docker image:
`docker run -it --rm gnuradio/gnuradio:latest`
2. Use a Signal Source block (or `analog.sig_source_c` in Python) to generate simulated IQ samples - no antenna or SDR device needed.
3. Run your script and verify the console output shows:
 - Correct detection when signal power is above threshold
 - Correct non-detection when power is below threshold
 - Correct classification (analog vs digital) for each signal type

4. To quickly verify the power measurement, compare your output against a known reference: a 0 dB full-scale sinusoid from sig_source should read ~0 dBFS.

Evaluation Criteria

- Code runs without errors from a clean install
- Power measurement logic is correct and explained
- Classification method is justified (doesn't have to be perfect, but must be reasoned)
- README is clear enough that someone unfamiliar with the code can run it
- Code is readable: meaningful variable names, at least brief comments on non-obvious parts